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Effect of different rotation systems on mercury methylation in paddy fields

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ABSTRACT

Rice (*Oryza sativa*) paddy is the hotspot of mercury (Hg) methylation. Given distinct influences of rotation systems on the physicochemical properties of paddy soils, we hypothesized different rotation systems in rice paddies inducing a large difference in Hg methylation. Here, we investigated Hg species distribution, dissolved organic matter (DOM) features, and Hg methylation in five rotation systems (Other farmland newly reclaim into paddy field, i.e., NR-R; Drain the water in winter, i.e., DW-R; Flooding in winter, i.e., FW-R; Rape-Rice rotation, i.e., Ra-R; Wheat-Rice rotation, i.e., Wh-R) of paddy fields to identify such hypothesis. Results shown that FW-R had the strongest Hg methylation, followed by Ra-R and Wh-R, then DW-R, and finally NR-R. We further found that much higher soil organic matter (SOM) and organo-chelated Hg (Hg-o) from straw residues and root exudates were the main cause for the greater Hg methylation in FW-R, Ra-R and Wh-R. This was because the protein-like fraction of SOM facilitated the net production of methyl Hg (MeHg), meanwhile the humin-like fraction had a strong affinity to MeHg in paddy soils. Therefore, it can be concluded that paddy soil under DW-R was the optimum pattern in order to reduce the occurrence of Hg methylation. However, paddy soils under Ra-R and Wh-R were the recommendable patterns if the productivity of paddy fields was considered.

1. Introduction

Rice (*Oryza sativa*) is a staple food crop, and its cultivated area reaches up to 0.143 billion hectares worldwide. Investigations indicated that approximately 0.48 billion tons of milled rice annually produced to feed about 3.5 billion populations around the world (Muthayya et al., 2014). In China, milled rice fed over 60% population with an annual production of 0.14 billion tons, accounting for 29.9% of the global total production in 2017 (USDA, 2017). Recent studies found that the rice grain could strongly accumulate MeHg (Hua et al., 2010; Meng et al., 2011, 2014; Qiu et al., 2008), a neurotoxic heavy metal pollutant (Clarkson, 2002), posing a high MeHg pollution risk in China, especially in Hg contaminated area (such as Hg mining area) (Qiu et al., 2008). Hence, consuming rice had been considered as the important pathway for MeHg exposure to residents in China.

It has been well documented that diverse rotation systems in paddy fields make distinct differences in physicochemical properties (Zhou et al., 2014) of paddy soils, such as the chemical speciation, biological effectiveness of heavy metal, DOM components, the diversity of microbial communities (Navarro-Noya et al., 2013) and enzyme activities (Janzen et al., 1992). Earlier studies also suggested that the long-term flooding in paddy field ecosystems promotes the Hg methylating microbes to induce Hg methylation rate (Hsu-Kim et al., 2013; Lin et al., 2014), leading to a high level of net MeHg production in paddy soils (Hamdy and Noyes, 1975; Hsu-Kim et al., 2013; Tanner et al., 2018). MeHg accumulation in rice grain is significantly positive correlation with the MeHg level in paddy soil (Meng et al., 2011; Qiu et al., 2008), thereby long-term flooding in paddy soils distinctly enhances the MeHg absorption by rice roots. It is noteworthy that the traditional planting patterns of rice in China was dominated by such flooded cultivation, thus posing a high risk of MeHg pollution. We also noticed there were other rotation systems in different regions of China due to the regional climate and terrain influences and plant customs of farmers. Basing on large scale surveys in Chongqing, Southwest of China, we found five rotation systems in paddy fields as: other farmland newly reclaim into paddy field (NR-R), drain the water in winter (DW-R), flooding in winter (FW-R), rape-rice rotation (Ra-R), wheat-rice rotation (Wh-R), respectively. Given impacts of rotation systems on

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paddy soil physicochemical properties, we hypothesized different rotation systems in paddy fields inducing a large difference in Hg methylation.

Low Hg methylation rotation system with a high economic value in paddy field is largely preferred in China. In this study, we performed a systematic investigation of paddy soils in Chongqing in 2017. Up to 74 paddy fields were investigated and 222 paddy soils were collected to evaluate the Hg methylation in above five rotation systems. We further analyzed total Hg (THg) concentration, MeHg concentration, Hg species distribution, soil pH, pore water Eh, SOM and DOM in these paddy soils to understand the governing factors for the difference of Hg methylation in these rotation systems. Finally, a suitable pattern is recommended to the farmer to reduce the exposed risk of MeHg from consuming rice.

2. Methods

2.1. Study area

Chongqing is a typical rice planting province in the southwest China (Fig. 1), and farmland accounts for 19.1% of land area. It belongs to the mid-subtropical humid monsoon climate and is located in the subtropical region. The temperature of Chongqing ranges from -8.1°C to 43.6°C with an annual average of 18.1°C. The rice growing period in the study region usually starts at early May and end up at the end of September, with temperature and humidity varying from 19.7°C to 42.2°C, and 62.6 to 96.1%, respectively. In the present study of 2017, approximately 59.5-70.3% of the annual rainfall occurs during the rice growing period (from May to September).

As one of the main megacities in southwest China, Chongqing presents a wide variation in soil Hg concentrations (1-6.80 × 10^4 ng g^{-1}) (Shi et al., 2013; Xu et al., 2018). Several mines (Hg and Mn) distribute in the southeast Chongqing, where is adjacent to the Wanshan Hg mine, Guizhou (a famous Hg mine in China). In the non-mine area, Hg contaminants are mainly from anthropogenic sources, such as industrial manufacture, fossil fuels combustion and residential emissions. Atmospheric Hg deposition in the southeast and urban area (Fig. 1) are higher than other areas. The agricultural areas mainly distribute in the central and western regions.

2.2. Sampling approach and pretreatment

Field investigation was performed from late April and early May 2017 when the paddy fields were flooded for preparing to transplant rice seedlings. A total of 222 soil samples were collected from 74 paddy fields (Table S1), which distributed in 33 districts in Chongqing (Fig. 1). All sampling paddy fields were away from highways, factories, cities and mines. Three soil samples were randomly collected via a cylindrical hand corer in each sample field, with approximately 150 g of the topsoil (0–20 cm) being collected for the subsequent analyzing. All soils were enclosed in individually labeled polyethylene (PE) ziplocked bags then stored in a car refrigerator (4°C) to transport to the laboratory within 48h. To avoid the influence of soil types, only the purple soil, which was the main soil type in Chongqing, was selected. Meanwhile, five typical rotation systems were classified in this study (Table 1).

All samples were freeze-dried (ZX-LGJ-18, Zhixin Ltd., China) in the

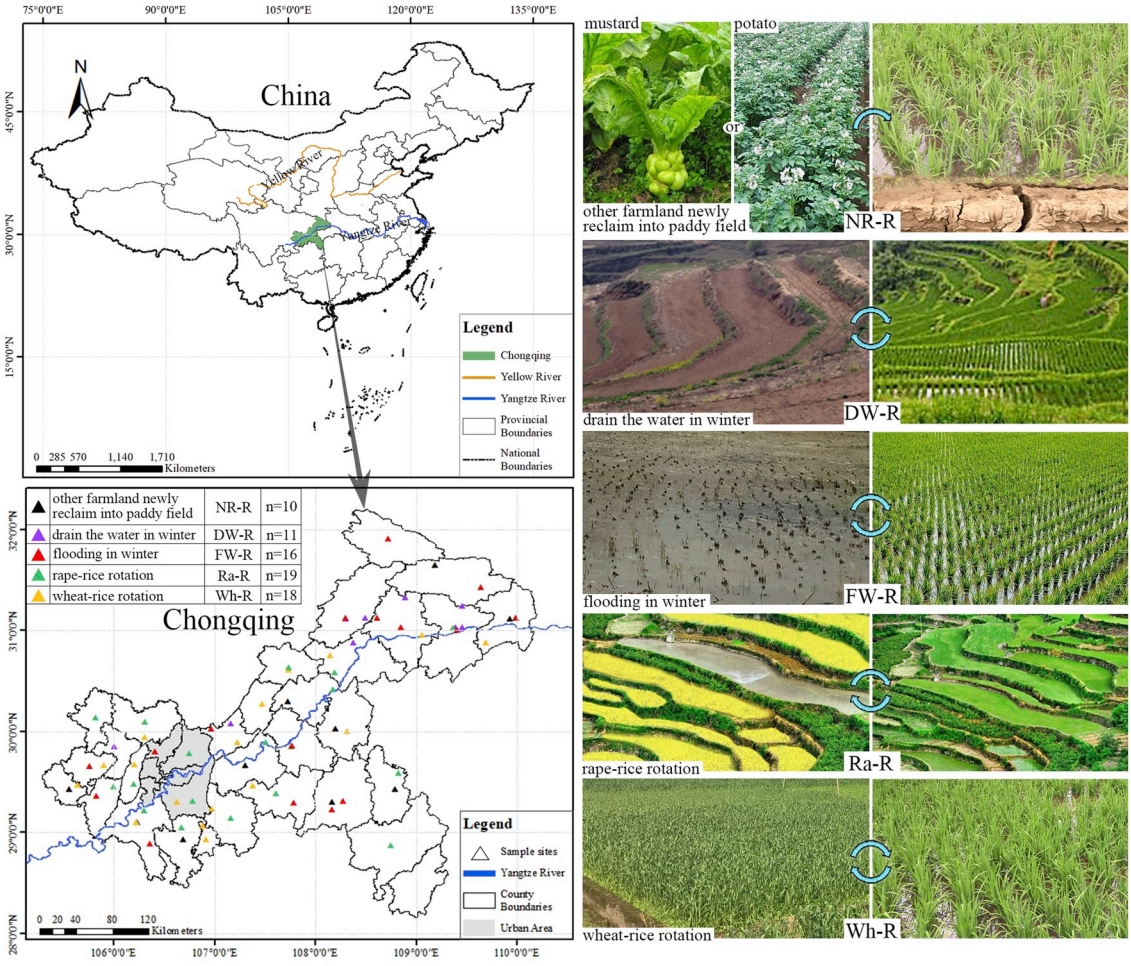


Fig. 1. Location and information of the study area and sampling sites.

Table 1
The classification of paddy fields.

Type	Classification	Abbreviation	Description
Single-cropping paddy field	Other farmland newly reclaim into paddy field	NR-R	Croplands that have been cultivated other crops for a long time, were newly reclaimed as paddy fields in rice planting season (from April to September).
	Drain the water in winter	DW-R	Paddy fields had drained the water to sun fields after rice harvest until the next rice planting season.
	Flooding in winter	FW-R	Paddy fields remained flooded with water after rice harvest until the next rice planting season. Lots of rice straw residues were discarded in these paddies.
Double-cropping paddy field	Rape-Rice rotation	Ra-R	Paddy fields were utilized to cultivate the rape from October and April.
	Wheat-Rice rotation	Wh-R	Paddy fields were utilized to cultivate the wheat from October and April.

laboratory. Then, sundries, such as straw residues and pebbles, were cleared away before grinding. All soil samples were ground in an agate mill and sieved through 150 meshes. The agate mill and sieve were required to clean by ultrapure water (18.2 MΩ·cm @25°C, Millipore, USA) and dry in a clean drying oven for each soil sample. During the above processes, disposable PE gloves were necessary to prevent any cross-contamination. All processed soil samples were stored in a -80°C freezer (DW-HL538, ZKmeiling Ltd., China) before further analysis.

2.3. Hg species analysis

Total Hg in soil was measured by a Direct Mercury Analyzer (DMA-80, Milestone srl., Italy) with a detection limit of 0.5 pg. The species of Hg in soil was performed using a modified method of sequential selective extractions (SSEs) (Bloom et al., 2003; Shi et al., 2005). Five fractions were separated through this method, as water-soluble Hg (Hg-w), 'human stomach acid' soluble Hg (Hg-h), organo-chelated Hg (Hg-o), elemental Hg (Hg-e) and mercuric sulfide (Hg-s), respectively. The detail extraction processes were showed in Table S2 (Supplementary material). The THg concentration of extracting solution in each above step was determined by cold vapor atomic fluorescence spectroscopy (CVAFS, Brooks Rand model III, Brooks Rand Laboratories, Seattle, WA, USA) (Wang et al., 2014), referring to EPA 1631 (EPA, 2002). For recoveries evaluation, THg concentrations of above five steps were summed and then compared to the corresponding THg concentrations measured by DMA-80. The error was controlled within 10%.

For soil MeHg analysis, approximately 0.50 g soil of each sample was accurately weighed and then digested with 1.5 ml $CuSO_4$ (2 mol l^{-1}) and 7.5 ml HNO_3 (25%, V/V) in an individual PE centrifuge tube for 60mins. Subsequently, accurate 10 ml dichloromethane (CH_2Cl_2) was added into the centrifuge tubes and closed with clean lids. Next, these centrifuge tubes were fixed to an orbital shaker for 160mins with a rotational speed of 200 r min^{-1}. Then, the substratum digestion solution was transferred to a new centrifuge tube and settled in a water bath (50°C) until the visible CH_2Cl_2 volatilized absolutely. Finally, the extracting solution was obtained behind removing residual CH_2Cl_2 with high purity N_2 at a water bath (75°C) for 10mins. MeHg in extracting solution was measured by an Automated Methyl Mercury Analytical System (MERX) (Brooks Rand Labs, Seattle, WA, USA), referring to the EPA method 1630 (EPA, 1998).

2.4. Soil DOM, SOM and pH

DOM was extracted by adding 45 ml of deionized water to 4.5 g soil in a 50 ml centrifuge tube (Ohno et al., 2007). Then, these closed centrifuge tubes were shaken on an opaque orbital shaker for 16h at room temperature, centrifuged at 1500 × g for 40mins, and filtered through a 0.45 μm cellulose acetate filter membrane (Aiken et al., 2011; Bolan et al., 2011; Jiang et al., 2017). The obtained extracted solution was stored in a brown bottle at 4°C. In particular, all subsequent works had to be completed within one week. DOM concentrations of extracting solution were represented by the dissolved organic carbon (DOC) concentrations (Jiang et al., 2018), which were determined by a Sievers InnovOx Laboratory TOC Analyzer (Detection limit, 0.05 mg l^{-1}).

The dichromate redox method (Carter and Gregorich, 2007) was used to measure SOM. Soil pH was determined with distilled water at a ratio of 1g:2.5ml with a pH meter (PB-10, Sartorius, Germany). When field sampling, pore water redox potential (Eh) was monitored at a depth of 5 cm in each paddy soil with a portable Eh meter (IQ-160, IQ Scientific instruments Co., USA) and ISFET-tipped electrodes inserted into the soil until the reading was stable.

2.5. DOM structure

Ultraviolet-Visible absorption and Excitation-Emission-Matrices (EEMs) spectra were obtained using an Aqualog® spectrophotometer (Horiba, France), which provided light via a 150 W ozone-free xenon lamp and worked at constant room temperature (20 ± 3°C). In Brief, the setting of the Aqualog® spectrophotometer and methods of determining UV-VIS absorption spectra and EEMs of fluorescence spectra were referred to Jiang et al. (2017).

Four indices were used to analyze the DOM structure, they were α(355) (the absorption coefficient at 355 nm wavelength), fluorescence index (FI), humification index (HIX), biological index (BIX), respectively. The α(355) was generally used to represent the relative concentration of chromophoric DOM (CDOM) (Hong et al., 2005; Miskimmin, 1991), while the FI can identify the autochthonous (i.e., microbial/algae) and allochthonous (i.e., terrestrial) sources in corresponding to FI > 1.9 and FI < 1.4, respectively (Huguet et al., 2009; McKnight et al., 2001; Wickland et al., 2007). As a surrogate of FI, the BIX can indicate the recently yielded autochthonous DOM and the contribution of protein-like substances (Wilson and Xenopoulos, 2008). Furthermore, HIX represents the humification of DOM, therefore, the high HIX values are generally associated with the high humification and structural complexity (Ohno et al., 2007). Their formulas are showed as follow:

$$\alpha_{355} = \frac{2.303 \times A_{355}}{r} \quad (1)$$

$$FI = \frac{f_{(Ex=254nm,\ Em=470nm)}}{f_{(Ex=254nm,\ Em=520nm)}} \quad (2)$$

$$HIX = \frac{\int (Ex = 254nm,\ Em = 435 - 480nm)}{\int (Ex = 254nm,\ Em = 300 - 345nm)} \quad (3)$$

$$BIX = \frac{f_{(Ex=310nm,\ Em=380nm)}}{f_{(Ex=310nm,\ Em=430nm)}} \quad (4)$$

where A355 is the absorbance at 355 nm wavelength; r is the length of the light path (0.01 m); f is the fluorescence intensities at a certain wavelength; Ex is excitation wavelength (nm); Em is emission wavelength (nm).

The details of spectral parameter calculations refer to Jiang et al. (2017).

2.6. Statistic analysis

Data management and correlation analysis were carried out using SPSS 21.0 (IBM, USA). In addition, One-way analysis of variance (ANOVA) was used to judge the differences between different data groups (Stephen, 2016). Pearson's correlation coefficient (r) was used for correlation analysis, whilst significance levels (p-value) were used to characterize the degree of correlation, $0.01 < p < 0.05$ is considered as "statistically significant", and $p < 0.01$ is "extremely significant" (Nuzzo, 2014).

2.7. Quality assurance (QA) and quality control (QC)

In laboratory analysis, ultra-clean bench, water (18.2 MΩ·cm @ 25°C, Millipore, USA) and reagents (GR) were requisite to avoid contamination and ultra-low blanks. All vessels were selected for and connecting pipes for Teflon, preventing adsorption from vessels and pipes. Ultrapure water was used as blank in Uv and EEMs spectra determining. The carrier gas was high purity N_2 (99.999%), Ar (99.999%) and O_2 (99.999%) for separation in CVAFS, Brooks Rand model III and DMA-80, respectively.

Method blanks, duplicates, matrix spikes, and certified reference materials (CRMs) were the essential processes for quality assurance and quality control. The deviations of replicate measurements were less than 10% for THg and 5% for MeHg. The standard substances (GBW07428/GSS-14, THg: 89 ± 4 ng g^{-1}) were used to calibrate DMA-80 and recover matrix spikes in CVAFS. The TORT-2, a certified reference material (National Research Council Canada, Lobster Hepatopancreas, MeHg: 152 ± 13 ng g^{-1}), was used as the standard material for MeHg determining. Recoveries for matrix spikes ranged from 90 to 110% for THg and from 85 to 105% for MeHg.

3. Results and discussion

3.1. THg and MeHg in paddy soils of different rotation systems

Large variations of THg concentrations were observed in paddy soils, ranging from 24.0 to 275 ng g^{-1} with a mean of 80.0 ng g^{-1} (Table S3 Supplementary material). The concentrations of THg in all the soil samples did not exceed the standard Hg level for safe agriculture soil (500 ng g^{-1}) recommended by Chinese government (HJ/T332-2006) (Agency, 2007) as well as the probable effect concentration of 1060 ng g^{-1} (MacDonald et al., 2000), demonstrating no significant Hg pollution risk in this study region. As shown in Fig. 2A, the highest average of THg concentrations was observed in Ra-R, and the minimum average in DW-R. There was, however, no significant difference ($p > 0.05$) among these five patterns, suggesting that the effect of difference of rotation systems on soil THg concentrations was not obvious. This maybe ascribe the large range of THg concentrations in the paddy soils of Chongqing and the random selection of sample paddy fields (Fig. 1).

Methyl Hg in paddy soils varied between 0.23 and 4.26 ng g^{-1} with an average of 0.96 ng g^{-1} (Table S3 Supplementary material), and 67.6% of sample soils were lower than this average value. For different rotation systems (Table 2B), the average of MeHg concentrations peaked in FW-R (1.80 ± 1.04 ng g^{-1}), while the minimum average in NR-R (0.41 ± 0.19 ng g^{-1}) (Table S3 Supplementary material). The averages of MeHg concentrations in Ra-R (1.00 ± 0.53 ng g^{-1}) and Wh-R (0.74 ± 0.30 ng g^{-1}) were lower than that in FW-R, but higher than those in DW-R (0.44 ± 0.15 ng g^{-1}) and NR-R (0.41 ± 0.19 ng g^{-1}) (Table S3 Supplementary material). The MeHg level in FW-R was significantly ($p < 0.05$) higher than those in other rotation systems. No significant difference ($p > 0.05$) was observed between DW-R and NR-R, as well as between Ra-R and Wh-R. These results suggested that the pattern of FW-R was prone to accumulate high level of MeHg.

The ratios of MeHg over THg concentrations (MeHg/THg) in paddy soils ranged from 0.37 to 3.67%, with a mean of 1.19%. Particularly, the highest MeHg/THg ratio (3.67%) was observed in a FW-R with a THg concentration of 81.5 ng g^{-1} (pH, 6.06; Eh, -201 mV), which was close to the THg average level (80.0 ng g^{-1}) for all paddy soils. This implied that not all Hg in paddy soil was involved in Hg methylation process. The reason probable was the Hg species was another important factor to shape the Hg methylation by Hg-methylating microorganism

Table 2
The concentrations of dissolved organic matter (DOM) and proportion of DOM over soil organic matter (SOM) in field investigation.

		NR-R	DW-R	FW-R	Ra-R	Wh-R
DOM (mg g^{-1})	Range	0.08-0.20	0.08-0.18	0.11-0.30	0.10-0.55	0.14-0.30
	Mean	0.11^b	0.12^b	0.18^a	0.20^a	0.19^a
DOM/SOM (‰)	Range	8.95-9.04	8.96-9.18	8.87-9.16	8.86-10.3	8.88-9.15
	Mean	8.96^a	9.03^a	8.99^a	9.09^a	9.03^a

Note: The same lowercase represents no significant difference ($p > 0.05$).

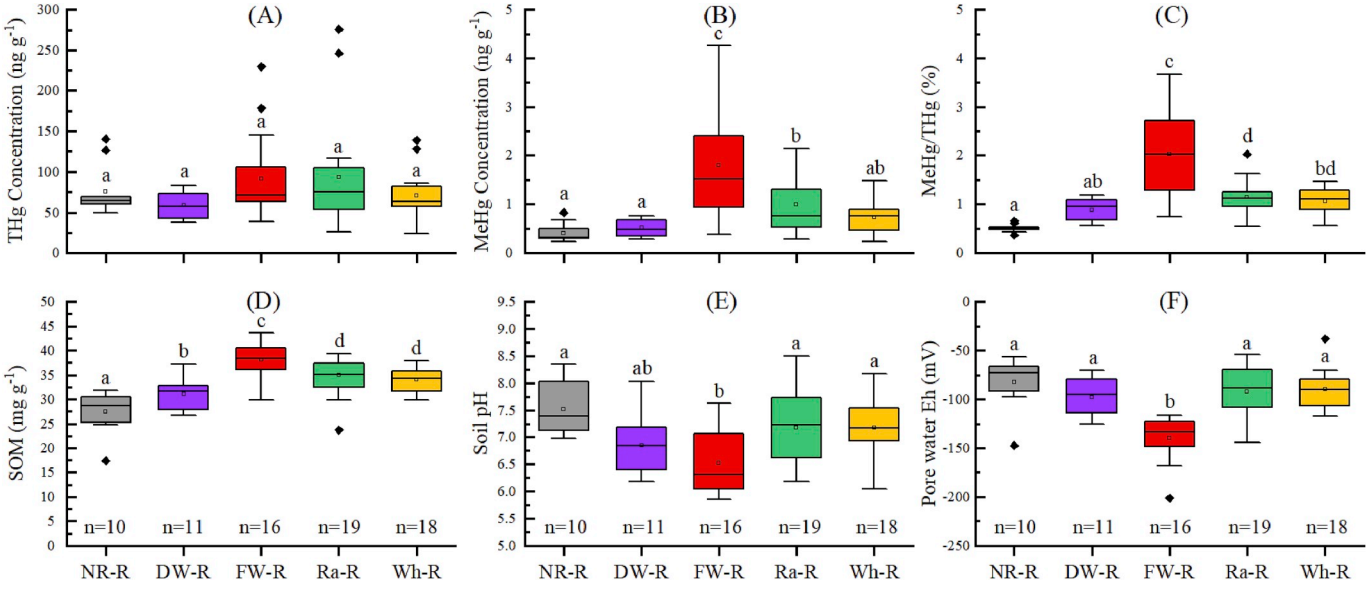


Fig. 2. Statistics of total mercury (THg) concentrations, methyl mercury (MeHg) concentrations and MeHg/THg ratios in paddy fields with different rotation systems (The same lowercase represents no significant difference ($p > 0.05$)).

(Shi et al., 2005). For different rotation systems, FW-R emerged the highest MeHg/THg average (2.04 ± 0.86%), and NR-R presented the lowest average (0.51 ± 0.08%). The average of MeHg/THg ratios in FW-R was 2-4 times ($p < 0.05$) higher than those in other rotation systems. Additionally, the averages of MeHg/THg ratios in NR-R (0.51 ± 0.08%) and DW-R (0.75 ± 0.19%) were lower than those in Ra-R (1.15 ± 0.33%) and Wh-R (1.07 ± 0.26%), and the average of MeHg/THg ratios in NR-R was significantly ($p < 0.05$) lower than those in Ra-R and Wh-R.

From the above statistics analysis, it was shown that Hg methylation in paddy soils was stronger than that in other agricultural soils, which cultivated other crops (such as mustard and potato). Hg methylation in the pattern of FW-R was the strongest, followed by Ra-R and Wh-R, while DW-R and NR-R were the weakest. Therefore, in terms of receding the risk of MeHg exposing, DW-R maybe the optimum rotation system for the paddy field. However, combining the productivity of the paddy field, we suggest the Ra-R or Wh-R was the optimal rotation system for paddy fields since with high economic value and low risk of MeHg pollution.

3.2. Influences of SOM, soil pH and pore water Eh on Hg methylation

The variation ranges of SOM, soil pH and pore water Eh were from 17.4 to 43.6 mg g^{-1}, 5.86 to 8.50 and -201 to -37.9 mV, respectively. The averages of SOM, soil pH and pore water Eh were 33.8 mg g^{-1}, 7.04 and -101 mV, respectively (Table S3 Supplementary material). All paddy soils shown reduced conditions because paddy fields had been flooded to prepare to transplant rice seedlings when the field investigation was conducted. It is noteworthy that FW-R had the longest flooding time, and NR-R, DW-R, Ra-R and Wh-R had nearly one-month flooding time. This was the reason we observed higher Eh in NR-R, DW-R, Ra-R and Wh-R than that in FW-R.

The MeHg/THg ratios exhibited a significant positive correlation ($p < 0.01$, n = 74) to SOM, while a significant negative correlation ($p < 0.01$, n = 74) to soil pH and pore water Eh (Fig. S1 Supplementary material). We found the variation of SOM contrasted with soil pH and pore water Eh (Fig. 2D-F). The FW-R, which with the greatest Hg methylation, had the highest SOM level while the lowest Eh ($p < 0.05$). Additionally, the SOM in Ra-R and Wh-R were significantly ($p < 0.05$) higher than those in NR-R and DW-R. Nevertheless, soil pH and pore water Eh were comparable in above two type rotation systems ($p > 0.05$). Overall, we suggest the rotation systems have distinct impacts on SOM level in paddy soils.

The strongest Hg methylation, highest SOM and lowest pH and Eh values were found in the FW-R could be attributed to the decomposing of straw residues. The rice straw residues were discarded and remained in the field after the rice harvest. Then the long-time flooding enhanced the decomposition processes and led to a distinct increase of SOM, an additional source of Hg, a decline of soil pH, and an anaerobic condition in paddy soils, thereby greatly inducing elevated Hg methylation. In addition, the MeHg/THg ratios and SOM in Ra-R and Wh-R were higher than those in NR-R and DW-R. This could be attributed to the greater release of root exudates and the litterfall of crops when cultivated rape and wheat in Ra-R and Wh-R than in NR-R and DW-R. It is well known that organic acids, sugars and amino acids in root exudates have such important impacts on the decline of soil pH and stimulation of microbial activity, thereby promoting bioavailability of Hg and net Hg methylation (Bais et al., 2006; Liu et al., 2018). This was the cause we observed high ratios of MeHg/THg in Ra-R and Wh-R. The DW-R shown relatively low MeHg/THg ratio and SOM level, which could be attributed to sunning in winter (an agricultural management pattern of paddy field). Sunning in winter decreases of soil moisture and enhances oxygen, which change microbial communities and reduce the microbial activities (Tanner et al., 2018). In addition, the paddy soil sunning in winter can also increase soil porosity, which is beneficial to transform Hg(II) into elemental Hg (Hg^0) and then re-emit to the atmosphere (Ci et al., 2016). Therefore, the pattern of DW-R is one of the good patterns to reduce Hg methylation in paddy soils. Finally, the pattern of NR-R shown the lowest MeHg/THg ratio and SOM level. The pattern of NR-R was newly reclaimed farmland that had cultivated other crops for many years, such as mustard and potato. These farmlands had been exposed to oxidation or weak reducing condition for a long time, hence less prone to Hg methylation.

3.3. Influences of Hg species on Hg methylation

The transport and transformation of Hg in paddy soil are mainly controlled by Hg species (Stein et al., 1996). As shown in Fig. S2 (Supplementary material), Hg-w was the minimum fraction in all soil samples, whose proportions ranged from 0.46% to 4.40%. However, Hg-w has the highest environmental risk because of its strong chemical activity, it can be absorbed by organisms directly (Issaro et al., 2009; Shi et al., 2005). Hg-w also readily binds to some ligands (Schuster, 1991), such as mineral colloids, organic matter and sulfide, and then transforms into insoluble complexes (such as Hg-o, Hg-e, Hg-s) that cannot be utilized by organisms directly (Shi et al., 2005). Similarly, Hg-h also accounted for a low percentage (1.11-8.06%) of THg. Since Hg-h can characterize the Hg bioavailability inside organisms (Bloom et al., 2003), it is also regarded to have high environmental risk. Once Hg-h is absorbed by organisms, it is likely to be transformed into Hg(II) and even MeHg. The Hg-o fraction occupied between 11.2% to 39.0% in THg, which made an apparent enhancement in contrast to the bioavailable fractions (BF), such as Hg-w and Hg-h. Hg-o can be utilized by the organism directly or utilized after the decomposing of organic matter, it has an intimate association with the carbon biogeochemical cycle in paddy soils. Therefore, it has a high potential environmental risk. Relatively inert species, Hg-e and Hg-s, accounted for an absolute high proportion in paddy soils, with a proportion ranging from 52.6% to 89.6%. Hg-e and Hg-s are as the product from the processes that Mn oxides and sulfide actively absorbing or binding Hg(II) and then transforming into inert complexes (Lockwood and Chen, 1973; Lu et al., 2014), such as Hg-Mn-Fe mixture (Lu et al., 2014), Hg(II)-polysulfides (Liu et al., 2009). Such processes can be accompanied by the transformation from Hg-w to relatively inert fractions, thereby reducing the bioavailable of Hg and inhibiting the Hg methylation in paddy soils. Although the Hg-e and Hg-s maybe convert to other bioavailable fractions under changes of soil properties (such as changes of pH, the addition of modifier) (Liu et al., 2009; Schuster, 1991; Wang et al., 2012) and transform into MeHg by Hg-methylating microorganism, this process is very slow and inefficient (Bloom et al., 2003). Therefore, the potential environmental risks of these two inert species are low in paddy soils. Overall, our results suggested that the BF Hg in paddy soils just took up a small fraction of THg.

The correlation analysis (Table S4 Supplementary material) indicated that the MeHg/THg ratios showed significantly positive correlation with Hg-o/THg ratios (r = 0.621, $p < 0.01$, n = 74) and BF/THg ratios (r = 0.321, $p < 0.01$, n = 74). Contrastly, a significant negative correlation was observed between MeHg/THg and Hg-s/THg ratios (r = -0.584, $p < 0.01$, n = 74). Furthermore, the MeHg/THg ratios also exhibited a positive correlation with Hg-w/THg ratios (r = 0.291, $p < 0.05$, n = 74) and Hg-h/THg ratios (r = 0.264, $p < 0.05$, n = 74). These results suggested that Hg-w, Hg-h and Hg-o played the key roles in Hg methylation in paddy soils. As discussed above, Hg-w and Hg-h can be utilized and transformed into MeHg by Hg-methylating microorganism (Hamdy and Noyes, 1975; Hsu-Kim et al., 2013). Unlike Hg-w and Hg-h, most Hg-o (such as particle Hg-o) cannot be directly utilized by Hg-methylating microorganism (Ravichandran, 2004) and it contains most MeHg (Karlsson and Skyllberg, 2003), thus the Hg-o/THg ratios can also reflect the accumulation of MeHg in paddy soils indirectly.

Hg-o/THg ratios emerged the same tendency with MeHg/THg ratios in different rotation systems (Figs. 2C and 3D), while Hg-s/THg ratios

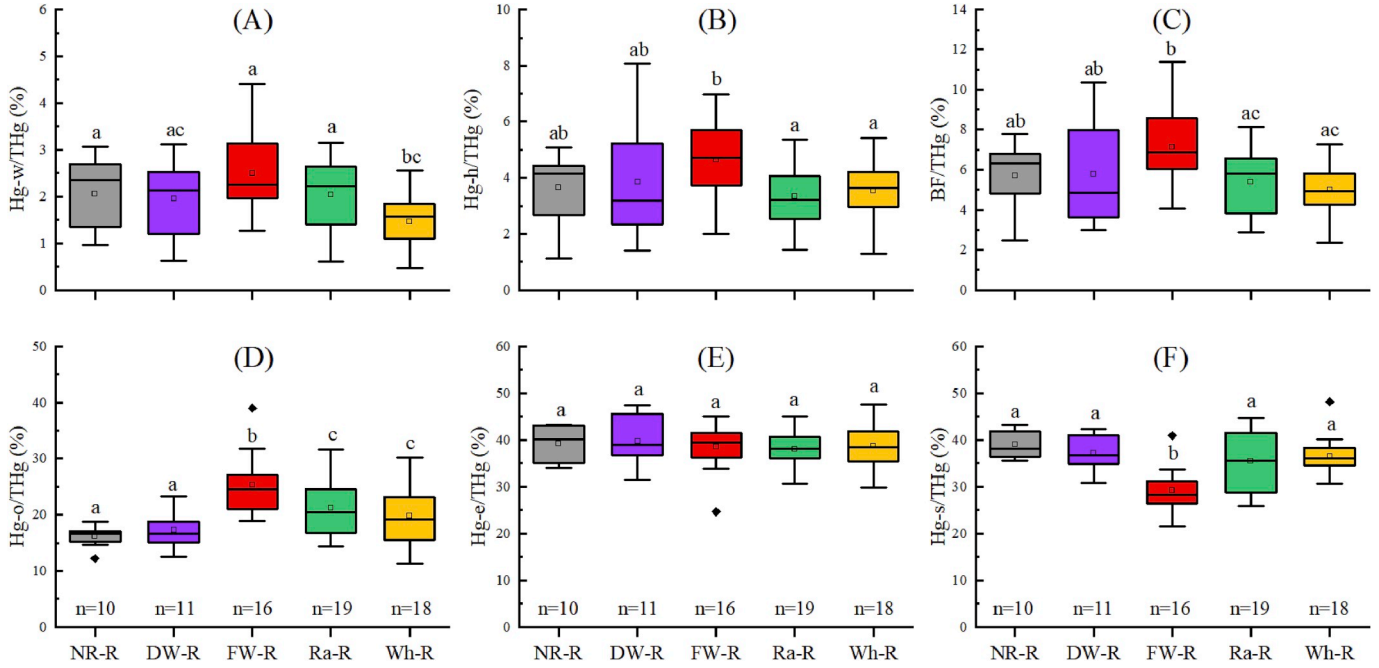


Fig. 3. Statistics of mercury (Hg) species to total mercury (THg) in paddy soils with different rotation systems (The same lowercase represents no significant difference ($p > 0.05$)).

showed an opposite tendency (Figs. 2C and 3F). Except that Hg-o/THg ratios and Hg-s/THg ratios in FW-R were significantly different ($p < 0.05$) with other rotation systems, no significant differences of other Hg species were observed between different rotation systems (Fig. 3). These results suggested the difference of rotation systems resulted in the divergence of Hg-o/THg ratios in paddy soils. In addition, the Hg-w/THg, Hg-h/THg, BF/THg and Hg-o/THg ratios in FW-R with the highest MeHg/THg level were higher than those in other rotation systems, while the lowest of Hg-s/THg level, implying that the relatively inert Hg species transformed into the bioavailable fractions under the condition of high SOM, low soil pH and pore water Eh during the long-time flooding. Meanwhile, high SOM in paddy soils easily leads to the high level of Hg-o/THg ratios due to the strong affinity between Hg and SOM (Hammerschmidt et al., 2008; Ullrich et al., 2001). This can be verified by the significant positive correlation between Hg-o/THg ratios and SOM (r = 0.373, $p < 0.01$, n = 74) (Table S4 Supplementary material).

The above analysis of Hg species in paddy soils highlighted the important role of Hg-o in Hg methylation in different rotation systems, consistent with earlier results (Seelen et al., 2018; Shi et al., 2005; Xu et al., 2017). The role of SOM on the Hg biogeochemical cycle highly depends on the SOM multiple sources and components (Bravo et al., 2017). First, the particle organic matter (POM) in soil/sediment plays a role on adsorbing Hg due to the high affinity between Hg and SOM (Hammerschmidt et al., 2008; Ullrich et al., 2001). The POM-Hg complexes (part of Hg-o) are not readily available to Hg-methylating microorganism (Ravichandran, 2004). Even the high POM level can not induce great level Hg methylation (Hammerschmidt et al., 2008). With the POM decomposition, the products of DOM and other complexes, such as the low-molecular-weight sulfur compounds (Cutter and Krahforst, 1988), can largely enhance MeHg producing rate in paddy soils (Bravo et al., 2017; Hsu-Kim et al., 2013; Zheng et al., 2012). Finally, the strong affinity between DOM and MeHg (Miskimmin, 1991) promotes the mobility of MeHg in paddy soils, thus inducing MeHg uptake by rice root.

3.4. Influence of DOM structure on Hg methylation

For different rotation systems, the highest average of DOM concentrations was observed in Ra-R, the lowest in NR-R (Table 2). The averages of DOM concentrations in Ra-R, Wh-R and FW-R were significantly higher ($p < 0.05$) than those in DW-R and NR-R, suggesting that Ra-R, Wh-R and FW-R could elevate DOM levels in paddy soils due to influences from root exudates and straw residues (Pinton et al., 2007). The lower DOM level in FW-R was caused by that nearly up to 8 months flooding time of FW-R led to substantial DOM runoff, thus left the bio-refractory such as cellulose and lignin (Devevre and Horwath, 2000) in paddy soils. It is noteworthy that the ratios of DOM over SOM (DOM/SOM) ranged from 8.86‰ to 10.3‰ with an average of 9.03‰ and these ratios were comparable in all rotation systems, indicating that the different rotation systems could not significantly affect the solubility of organic matter in paddy soils. Although DOM accounts for only a small part of SOM, it has significant impacts on the fate, bioavailability, transportation and transformation of Hg in the natural setting (Aiken et al., 2011; Bolan et al., 2011; Jiang et al., 2017).

As shown in Table 3, the tendency of CDOM was similar to that of

Table 3
Spectral parameters of dissolved organic matter (DOM) in soil extract from field samples.

Type	α(355) (m^{-1})	FI	BIX	HIX	(A + C + M)/Total (%)	T/Total (%)
NR-R	3.68 ± 1.51	1.47 ± 0.07	0.58 ± 0.03	3.46 ± 0.23	47.5 ± 9.19	4.48 ± 0.91
DW-R	4.92 ± 1.42	1.52 ± 0.06	0.59 ± 0.07	2.87 ± 0.30	50.2 ± 12.1	4.85 ± 1.27
FW-R	7.08 ± 4.20	1.45 ± 0.03	0.57 ± 0.03	3.13 ± 0.38	48.2 ± 9.95	3.47 ± 0.68
Ra-R	9.33 ± 2.76	1.63 ± 0.05	0.81 ± 0.07	2.80 ± 0.57	44.2 ± 3.18	6.09 ± 0.71
Wh-R	9.70 ± 3.68	1.59 ± 0.07	0.69 ± 0.08	2.83 ± 0.59	42.1 ± 2.68	5.16 ± 0.38

Table 4
The summarizes of different rotation systems paddy fields.

Abbreviation	Type	Sorting from high to low		Possible cause	Recommend
		MeHg (ng g^{-1})	MeHg/THg (%)		
NR-R	Other farmland newly reclaim into paddy field	0.41 ± 0.19	0.51 ± 0.08	Low SOM, Hg-o/THg, high pore water Eh, inert Hg species	
DW-R	Drain the water in winter	0.44 ± 0.15	0.75 ± 0.19	Low SOM, Hg-o/THg, high humin-like fraction, pore water Eh, inert Hg species	★
FW-R	Flooding in winter	1.80 ± 1.04	2.04 ± 0.86	Low soil pH, pore water Eh, inert Hg species, high SOM, bioavailable Hg species, Hg-o/THg	
Ra-R	Rape-Rice rotation	1.00 ± 0.53	1.15 ± 0.33	High SOM, protein-like fraction	★★
Wh-R	Wheat-Rice rotation	0.74 ± 0.30	1.07 ± 0.26	High SOM, protein-like fraction	★★★

Note: Hg-o/Hg means the ratio of organo-chelated mercury over total mercury. The amount of star symbol represents the recommended level. The cultivation of rape and wheat in winter will enhance the economic values of these paddy fields, hence we recommend Ra-R and Wh-R as the optimal rotation systems.

DOM in the five rotation systems, implying that Ra-R, Wh-R and FW-R could enhance CDOM levels in paddy soils. FI values of all soil samples were between 1.4 and 1.9, suggesting mixed source contributions for DOM in all kinds of paddy soils. In particular, FI of FW-R and NR-R were close to 1.4, suggesting significant allochthonous sources of DOM for FW-R and NR-R. Conversely, DOM in Ra-R, Wh-R and DW-R exhibited a relatively autochthonous feature. As an alternative of FI, BIX presented a positive correlation with FI ($r = 0.723$, $p < 0.01$, $n = 81$). The highest HIX was observed in NR-R, while the lowest in Wh-R, suggesting that DOM in Ra-R and Wh-R had relatively evident autochthonous characteristic and lower humification in contrast to other rotation systems. This is because the root exudates enhance the protein-like fluorophores of DOM due to its amino acid compositions (Pinton et al., 2007). Meanwhile, the root exudates are highly biodegradable and prone to be assimilated by microbes in the rhizosphere due to their high energy and carbon content (Pinton et al., 2007). The root exudates induce changes of soil characteristics (Pinton et al., 2007), which likely promotes the bioavailability of Hg and methylation rate by microbial metabolism (Janssen et al., 2016; Lin et al., 2014). DOM in DW-R, NR-R and FW-R showed stronger allochthonous characteristics and higher humification than those in Ra-R and Wh-R, which could be attributed to DOM with exogenous characteristics entering paddy soil with irrigation water and the highly humification of organic matter due to long-time flooding for FW-R, respectively.

Fluorescence peaks of A (Ex/Em = 250-260nm/380-480nm), C (Ex/Em = 330-350nm/420-480nm), M (Ex/Em = 315nm/437-441nm), and T (Ex/Em = 230nm/320-350nm) were observed in all soil samples, which represented humin-like fluorophores (including A, C and M) and protein-like fluorophores (T) (Kowalczuk et al., 2005; Parlanti et al., 2000). As shown in Table 3, the proportions of humin-like fluorophores were much bigger than those of protein-like fluorophores, implying that the humin-like fraction was further more than the protein-like fraction in DOM. The variation of T/Total ratios in different rotation systems was similar to BIX, which represented the contribution of protein-like fraction in DOM (Wilson and Xenopoulos, 2008). The T/Total ratios in FW-R were significantly lower ($p < 0.05$) than those in other rotation systems, while with higher (A + C + M)/Total ratios. This suggested that the DOM in FW-R had higher humification and lower bioavailability. Ra-R and Wh-R showed higher T/Total ratios in contrast to NR-R and DW-R, demonstrating that root exudates from rape and wheat in winter elevated the level of protein-like substances in paddy soils.

A significantly positive relationship was observed between (A + C + M)/Total and MeHg/THg ratios ($p < 0.01$, $n = 74$), while no significant correlation between T/Total and MeHg/THg ratios ($p > 0.05$, $n = 74$). Excluding the data of T/Total ratios in FW-R, the T/Total ratios had a significantly positive correlation with MeHg/THg ratios ($p < 0.01$, $n = 58$). Published studies indicated that the protein-like substances had high bioavailability (Wilson and Xenopoulos, 2008) and could facilitate the bacterial activity (Bravo et al., 2017), then enhanced the Hg methylation in paddy soils (Hsu-Kim et al., 2013). The humin-like substances with complex structure and long residence time were generally formed by secondary synthesis reactions of recalcitrant compounds, with products of microbial and chemical decay and transformation of biomolecules originated from organisms during life and after death (Bolan et al., 2011; Gaffney et al., 1996). Accordingly, MeHg bonds to humin-like substances when the protein-like substances stimulated bacteria to produce MeHg. As a result, MeHg accumulated in paddy soils with the accumulation of humin-like substances. In addition, the high MeHg concentrations and MeHg/THg ratios in FW-R were most likely attributed to the enhanced accumulation of the humin-like substances. The NR-R, DW-R, Ra-R and Wh-R were flooded for a short time, and their DOM was with high level of protein-like fraction. Such type of DOM had a close relationship with Hg methylation in paddy soils. This explained the significant correlation between T/Total and MeHg/THg ratios. Overall, the protein-like fraction in DOM played an important role in stimulating net Hg methylation in paddy soils, while the humin-like fraction played an important role in MeHg accumulation in paddy soils. These two fractions in DOM together determined the characteristics of Hg methylation in paddy soils.

4. Conclusions

As the summary in Table 4, this study shown that the rotation systems resulted in significant differences in SOM, Hg-o and MeHg/THg ratios. The protein-like fraction in SOM stimulated the net Hg methylation rate, and himun-like fraction in SOM promoted the MeHg accumulation in paddy soils. FW-R shown the highest Hg methylation, with 2-4 times higher than other rotation systems. DW-R was conducive to reduce Hg methylation in paddy soils. Ra-R and Wh-R showed relatively low Hg methylation rates. Combining the production of paddy fields, Ra-R and Wh-R were the recommended as the main rotation system in the paddy field because of with high economic value and low risk of MeHg pollution. Given protein-like substances facilitating the bacterial activity, we recommend further studies on polymerase chain reaction (PCR) amplification of mercury reductase and mercury methylase genes.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ecoenv.2019.109403.

Note: The α(355) represents the relative concentration of

chromophoric DOM. The FI identifies the autochthonous (i.e., microbial/algae) and allochthonous (i.e., terrestrial) sources in corresponding to FI > 1.9 and FI < 1.4, respectively. The BIX indicates the recently yielded autochthonous DOM and the contribution of protein-like substances. The HIX represents the humification of DOM. The (A + C + M)/Total and T/Total mean the percentage of humin-like substances and protein-like substances over total DOM.

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